Libicki, C. Martin (2009). Cyber Deterrence and Cyber War. Santa Monica, CA: RAND Corporation. pp. 23-74.

Richard J. Harknett. (1996). "Information Warfare and Deterrence".Parameters 26(3). pp. 93-107.

The common denominator in both articles is their recognition that traditional Deterrence strategies derived from nuclear or conventional military paradigms has little to no applicability to information warfare in general and "cyberwar" in particular. Libici declares upfront that one mustdifferentiate cyber deterrence from nuclear deterrence or general military deterrence as they all work to the detriment of cyber deterrence and in fact expose why and how cyberdeterrence is so problematic. Harknettconcursin saying that deterrence models developed during the Cold War will provide poor guidance for strategic thinking about this new form of war, which is better understood in the context of offense and defense.

Both articles however fail to position information (cyber) warfare where it belongs – an Intelligence covert warfare with no rules and no established norms and practices. In other words the field of inquiry is too new and has little to no research foundation and no empirical data. Both articles are prophetic in terms of elevating the topic to the forefront and identifying its problematic nature, yet its increased importance in future global and regional conflicts. The two articles were published in 2009 and 1996 (!) respectively and were likely dated shortly after their publication. The amazing technological advances in hardware, software, internet connectivity, surveillance tools and social networking are on a trajectory that makes academic study of the topic that much more difficult. Cyber war extends beyond military targets to unchartered areas such as finance, homeland security, foreign relations, strategic programs etc. (e.g., Israeli Credit Cards, Wikileaks, Iran's nuclear program). It is also misleading to apply the state-to-state paradigm in analyzing cyber warfare and especially deterrence strategies. There is no doubt that cyber war strategies and tactics (with the exception of hacking formischievous,fraud or profit motives)

are being used as an effective state sponsored weapons. Libici is right in pointing out that the admission, detection and attribution factors that are required to apply cost-benefit analysis to rational deterrence theories are not applicable in this case. It is reasonable to conclude that cyber warfare is still in its infancy, it has no rules or norms of warfare and it has no historical background from which to extrapolate methods of deployment. Yet it is in some ways the most dangerous form of warfare because of its unlimited potential to inflictEconomical, Military and Security damage in a surgical, clinical and largely undetectable manner.Most importantly unlike other forms of warfare it does not involve destroying physical infrastructure or human casualties.

Libi poses legitimate questions but fails to provide pragmatic tools and methods to contend with the issues. However highlighting the complexity of the factors and heightening the awareness of the phenomenon is a significant contribution in itself.

It is for these reasons that I strongly endorse Harknett's conclusion: *"The nature of netwar and cyberwar lend themselves to analytical frameworks and a strategic calculus dominated by offense-defense models, rather than by deterrence".*

As a post script, Yediot Achronot of 12-01-2012 reports that there appear to have been a MAD like tacit agreement struck between Israel and the Gulf States led by Saudi Arabia with regards to mutual de-escalation of "hacking" attacks and retaliations . According to the paper the reaction of the deputy Foreign Minister Dan Ayalon in response to the latest credit card hacking was interpreted as a declarion of "cybernetic war against Saudi Arabia"This is a clear indication that cyberwarfare has been legitimized as serious weapon on a state-to-state level. It may also indicate that Israel's well publicized internet technological prowess (Iran) have at least in this case acted as effective deterrence.